Sergio Arroni

AI Architect & Researcher

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/Sergio-arroni

https://sergioarroni.github.io/

/Sergio-Arroni-Del-Riego

About Me

AI Architect & Researcher specialized in taking AI solutions from research to production. Currently pursuing an Industrial PhD in Intelligent Systems, focused on generative AI for tabular data.

I work across the full AI system lifecycle: from problem definition and algorithmic design to production deployment and infrastructure management. I combine expertise in optimization, generative models, and MLOps to build scalable solutions with tangible business impact.

Papers

DOI: 10.1016/j.cose.2023.103546

DOI: 10.9781/ijimai.2023.02.005

Languages

Spanish · Native

English · Basic

Experience

AI Architect & Researcher

TheNextPangea S.L

Full-Time

Aug.. 2023 - Present

Aviles

- End-to-end AI system development: algorithmic design, Python implementation, and production deployment
- Design of optimization solutions using Bayesian Optimization (BoTorch) and evolutionary algorithms
- Research and development in generative AI for tabular data
- Development of RAG architectures with Gemini (GCP) for complex information processing
- Infrastructure and MLOps management on AWS with Terraform (IaC) and CI/CD
- Application of explainable AI techniques to enhance interpretability and adoption
- Technical coordination between research and engineering teams
- Participation in technical proposals and architecture definition for projects

Research Fellowship in Machine Learning

Fundación Universidad de Oviedo (FUO)

Full-Time

2021 - 2023

Oviedo

- Development of research projects in various fields (NLP, cybersecurity, and physics) applying Machine Learning techniques
- Design, training, and validation of models for real-world problems across different domains
- Publication of scientific articles in specialized journals
- Data analysis and experimentation for model and methodology evaluation

Education

2024 - Actualidad — PhD in Intelligent Systems
Universidad de Oviedo

2023 - 2024 — Master's in Artificial Intelligence
Universidad Politécnica de Madrid

2019 - 2023 — Bachelor's in Software Engineering
Universidad de Oviedo

Soft Skills

Knowledge Transfer

Impact- and Results-Oriented

Technical Decision-Making with Business Vision

End-to-End Autonomy and Ownership

Awards

2023 — Best Final Degree Project (TFG) at the University of Oviedo, focused on ICT topics
Awarded by Indra, Inetum, and Capgemini